

Figure 3: Overview of Cloudify architecture

application to scale automatically when the workloads increase.

Currently, only Java and Python are supported as the programming languages; however, GAE does provide an experimental API called 'Go', which enables users to program using additional languages such as Groovy, Scala and PHP.

Similar to Windows Azure, GAE also provides additional means for storing your data using three main services:

- **App Engine Datastore:** A NoSQL object store that comes bundled with a query engine.
- **Google Cloud SQL:** A highly scalable relational SQL database service that is very similar to MySQL.
- **Google Cloud Storage:** A simple storage container that can store files and objects pertaining to your application.

GAE is priced similar to Amazon's Elastic Beanstalk. It is a free utility. You only have to pay for the resources that you consume.

Cloudify: Cloudify is another example of an open source PaaS platform. It relies on a 'recipe-based' model to deploy applications on any cloud-based platform without having to undergo changes in the application code. The 'recipe' can be thought of as a template or blueprint for installing, configuring and monitoring the entire application stack. These 'recipes' are designed to be cloud-agnostic and, hence, can be executed on multiple cloud platforms without any major changes made to them.

Cloudify currently supports and provides recipes for a variety of application servers such as Tomcat, JBoss and Oracle Web Logic; for scalable databases such as Cassandra and MongoDB, and even for a host of private and public clouds including Amazon Web Services, Openstack, Cloudstack, etc.

Like most PaaS providers, Cloudify also provides automated auto-scaling and dynamic load balancing of resources.

Open Shift: Open Shift is a PaaS offering from Red Hat and it primarily comes in two variants—an enterprise PaaS that is meant to be deployed on private clouds, and the Web-

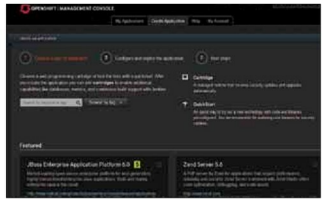


Figure 4: Open Shift management console

based Open Shift that allows you to launch your applications on a Red Hat provided cloud. There is an open source version available on GitHub as well, called Open Shift Origin.

Getting started with Open Shift is fairly easy. Developers can create an application using Open Shift's command line console or an IDE, code the application and then push it on to Open Shift.

Once published, Open Shift will take care of the application's underlying services and even scale the application automatically in case the workloads increase.

Open Shift supports a variety of development languages such as Java, Ruby, PHP, Python, Perl, Node.js as well as databases such as MySQL, PostgreSQL and MongoDB.

The Open Shift platform also provides a facility for continuous integration of code and its release management, using Jenkins. Once the application code is submitted, Jenkins automatically builds it, performs testing on the code and publishes it, depending on the test results. This enables you to develop applications in a more automated and streamlined fashion.

There are many more leading PaaS providers in the market today—Salesforce.com, Heroku, AppFrog, AppScale, etc—each one providing a different set of services and tools for developers to build and host their applications on. Their growing popularity proves that PaaS is a faster, cheaper, more reliable and secure way to develop applications, and is here to stay. **END** 

References

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